

MODULE 5

USER MANAGEMENT & AUTHENTICATION MODULE

5.1 Introduction

The User Management and Authentication Module is responsible for providing secure access to the Smart Event Management and Ticket Booking System. It enables users to create accounts, log in securely, and access features based on their roles. The module also provides separate authentication for administrators, ensuring that administrative operations remain protected from unauthorized access.

The authentication system has been developed using JSON Web Tokens (JWT), bcrypt password encryption, and MongoDB Atlas for secure storage of user credentials.

5.2 Objectives

The objectives of the User Management and Authentication Module are:

- To provide secure user registration.
 - To authenticate users before granting access to the system.
 - To implement separate authentication for administrators.
 - To ensure secure password storage using encryption.
 - To protect sensitive routes using JWT authentication.
 - To maintain user sessions securely.
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5.3 User Registration

The registration module allows new users to create an account by providing their personal details such as name, email address, and password.

During registration:

- User information is validated.
- Duplicate email addresses are prevented.
- Passwords are encrypted using bcrypt.
- User information is stored securely in MongoDB Atlas.

After successful registration, users can log into the system using their credentials.

5.4 User Login

The login module verifies the user's email address and password before granting access to the application.

The login process includes:

- User credential verification.
 - Password comparison using bcrypt.
 - JWT token generation.
 - Secure session creation.
 - Redirection to the user dashboard after successful authentication.
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5.5 Administrator Login

A separate authentication mechanism has been implemented for administrators.

Administrator login provides access to administrative functionalities while preventing normal users from accessing restricted features.

Administrator authentication includes:

- Secure admin login.
 - Role verification.
 - Protected administrator dashboard.
 - Restricted administrative privileges.
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5.6 User Dashboard

After successful authentication, users are redirected to their dashboard.

The dashboard provides:

- User information.
- Navigation options.
- Booking-related features.
- Event browsing.
- Secure logout functionality.

Protected routing ensures that only authenticated users can access dashboard pages.

5.7 Security Features

To improve system security, several authentication mechanisms have been implemented.

These include:

- Password hashing using bcrypt.
- JWT-based authentication.
- Role-based authorization.
- Protected API endpoints.
- Secure session handling.
- Environment variable configuration.
- CORS protection.

These mechanisms ensure secure communication between the frontend, backend, and database.

5.8 Database Integration

MongoDB Atlas is used for storing user information securely.

The database stores:

- User Name

- Email Address
- Encrypted Password
- User Role
- Account Creation Date

Database connectivity is managed using Mongoose.

5.9 API Implementation

The following REST APIs were implemented for authentication:

API	Method	Description
<code>/api/auth/register</code>	POST	Register a new user
<code>/api/auth/login</code>	POST	User login
<code>/api/auth/admin/login</code>	POST	Administrator login
<code>/api/auth/me</code>	GET	Retrieve logged-in user information
<code>/api/health</code>	GET	Backend health status

5.10 Testing

The authentication module was tested successfully.

The following test cases were performed:

- User Registration
- Duplicate User Validation
- User Login
- Invalid Login Credentials
- Administrator Login
- JWT Token Generation
- Protected Route Access
- Logout Functionality

- Database Connectivity
- API Testing

All test cases produced the expected results.

5.11 Deployment

The completed authentication module was successfully deployed to cloud platforms.

Deployment details:

Frontend

- Vercel

Backend

- Railway

Database

- MongoDB Atlas

Version control was managed using Git and GitHub, enabling continuous deployment and project maintenance.

5.12 Module Summary

The User Management and Authentication Module successfully provides secure access control for the Smart Event Management and Ticket Booking System. The implementation includes user registration, user login, administrator login, encrypted password storage, JWT-based authentication, protected routes, and secure API communication. The module has been integrated with MongoDB Atlas for cloud-based data storage and deployed using Vercel and Railway, ensuring reliable and secure access to the application.